

Cs401

Assembly Language

Most Repeated Subjective Midterm

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Short Question

Question No 1

Why is it necessary to provide the segment and offset address in case of FAR jump ?

Answer:-

Far jump must be used a two byte segment and a two byte offset are given to it Because It loads CS with the segment part and IP with the offset part.

Question No 2

What's your understanding about Incrementing and Decrementing Stack?

Answer:-

A decrementing stack moves from higher addresses to lower addresses as elements are added in it while an incrementing stack moves from lower addresses to higher addresses as elements are added.

Question No 3

What is difference between SHR and SAR instructions?

Answer:-

The shift logical right operation inserts a zero from the left and moves every bit one position to the right and copies the rightmost bit in the carry flag.

Question No 4

Define implied operand?

Answer:-

An implied operand means that it is always in a particular register say the accumulator, and it need not be mentioned in the instruction.

Question No 5

For what purpose "INT 1" is reserved?

Answer:-

INT 1, Trap, Single step Interrupt This interrupt is used in debugging with the trap flag. If the trap flag is set the single Step Interrupt is generated after every instruction. By hooking this interrupt a debugger can get control after every instruction and display the registers etc. 8088 was the first processor that has this ability to support debugging

Question No 6

Define short jump

Answer:-

If the offset is stored in a single byte as in 75F2 with the opcode 75 and operand F2, the jump is called a short jump.

Question No 7

Explain the function of rotate right (ROR) instruction

Answer:-

In the rotate right operation every bit moves one position to the right and the bit dropped from the right is inserted at the left. This bit is also copied into the carry flag.

Question No 8

How many parameters a subroutine can receive through stack.

Answer:-

The maximum parameters a subroutine can receive are seven

Question No 9

Double working of Exchange instruction with help of an example.

Answer:-

Xchg ax, [bx+si+2] ; exchange ax with second number

Question No 10

Relation between RET and CALL is dependent or independent.

Answer:-

Technically they are independent in their operation but both the instructions are commonly used as a pair. The RET works regardless of the CALL and the CALL works regardless of the RET.

Question No 11

What is the difference between LES and LDS instruction?

Answer:-

LES loads ES while LDS loads DS. These instructions have two parameters, one is the general purpose register to be loaded and the other is the memory location from which to load these registers.

Question No 12

Write down the procedure to clear the selective bit.

Answer:-

The source operand is loaded with a mask containing one at positions which are retain their old value and zero at positions which are to be zeroed. The effect of applying this operation on the destination with mask in the source is to clear the desired bits. This operation is called masking.

Question No 13

Why REP prefix is generally not used with LODS instruction?

Answer:-

LODS is generally used in a loop and not with the REP prefix since the value previously loaded in the register is overwritten if the instruction is repeated and only the last value of the block remains in the register.

Question No 14

What is difference between REPE and REPNE?

Answer:-

REPE or REPZ repeat the following string instruction while the zero flag is set and REPNE or REPNZ repeat the following instruction while the zero flag is not set.

Question No 15

Describe Push and Pop with the help of an example.

Answer:-

The operations of placing items on the stack and removing them from there are called push and pop. Memory is like a shelf numbered as zero at the top and the maximum at the bottom. If a decrementing stack starts at shelf 5, the first item is placed in shelf 5; the next item is placed in shelf 4, the next in shelf 3 and so on

Question No 16

How many block processing instructions in 8088, just name the number of processing instructions?

Answer:-

There are just 5 block processing instructions in 8088. The five instructions are STOS, LODS, CMPS, SCAS, and MOVS called store string, load string, compare string, scan string, and move string respectively

Question No 17

What are the instruction use by assembly language for permanent diversion and temporarily diversion?

Answer:-

The instructions for permanent diversion in 8088 are the jump instructions, while the instruction for temporary diversion is the CALL instruction.

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Question No 18

How much string instructions are in assembly language? Write down all instructions.

Answer:-

STOS – Clearing the Screen

LODS – String Printing

SCAS -String Length

LES and LDS

MOVS -Screen Scrolling

CMPS – String Comparison

Question No 19

For what purpose DS and ES registers are generally used in the context of video memory.

Answer:-

Both DS and ES can be used to access the video memory. However we commonly keep DS for accessing our data, and load ES with the segment of video memory.

Question No 20

What colors are repeated by 0th, 1st and 2nd bits (from least significant side) in Attribute Byte?

Answer:-

Blue , green , Red

Question No 21

Describe the working of the CALL instruction with the reference of Stack.

Answer:-

During the CALL operation, the current value of the instruction pointer is automatically saved on the stack, and the destination of CALL is loaded in the instruction pointer. Execution therefore resumes from the destination of CALL.

Question No 22

Explain the use of TEST instruction.

Answer:-

The test instruction is used for bit testing. BX holds the mask and in every next iteration it is shifting left, as our concerned bit is now the next bit.

Long Question

Question No 1

What is the Difference between CALL and RET

Answer:-

The CALL instruction allows temporary diversion and therefore reusability of code. The word return holds in its meaning that we are to return from where we came and need no explicit destination. Therefore RET takes no arguments and transfers control back to the instruction following the CALL that took us in this subroutine.

Question No 2

Affected flag of AND operation

Answer:-

AND can be used to check whether particular bits of a number are set or not. Previously we used shifting and JC to test bits one by one. Now we introduce another way to test bits, which is more powerful in the sense that any bit can be tested anytime and not necessarily in order. AND can be applied on a destination with a 1-bit in the desired position and a source, which is to be checked. If the destination is zero as a result, which can be checked with a JZ instruction, the bit at the desired position in the source was clear.

Question No 3

Describe Local Variable?

Answer:-

Another important role of the stack is in the creation of local variables that are only needed while the subroutine is in execution and not afterwards. They should not take permanent space like global variables. Local variables should be created when the subroutine is called and discarded afterwards. So that the space used by them can be reused for the local variables of another subroutine. They only have meaning inside the subroutine and no meaning outside it.

Question No 4

Explain all characteristics of SCAS instruction.

Answer:-

SCAS compares a source byte or word in register AL or AX with the destination string element addressed by ES:DI and updates the flags. DI is updated to point to the next location. SCAS is often used to locate equality or inequality in a string through the use of an appropriate prefix. SCAS is a bit different from the other instructions. This is more like the CMP instruction in that it does subtraction of its operands. The prefixes REPE (repeat while equal) and REPNE (repeat while not equal) are used with this instruction. The instruction is used to locate a byte in AL in the block of memory. When the first equality or inequality is encountered; both have uses.

Question No 5

Explain divide overflow error.

Answer:-

If a large number is divided by a very small number it is possible that the quotient is larger than the space provided for it in the implied destination. In this case an interrupt is automatically generated and the program is usually terminated as a result. This is called a divide overflow error.

Question No 6

For what purpose "INT4" is reserved.

Answer:-

The overflow flag is set if the sign bit unexpectedly changes as a result of a mathematical or logical instruction. However the overflow flag signals a real overflow only if the numbers in question are treated as signed numbers. So this interrupt is not automatically generated but as a result of a special instruction INTO (interrupt on overflow) if the overflow flag is set. Otherwise the INTO instruction behaves like a NOP (no operation).

Question No 7

What is the purpose of following string instructions?

STOS

CMPS

Answer:-

STOS:-

STOS transfers a byte or word from register AL or AX to the string element addressed by ES:DI and updates DI to point to the next location. STOS is often used to clear a block of memory or fill it with a constant.

CMPS:-

CMPS subtracts the source location DS:SI from the destination location ES:DI. Source and Destination are unaffected. SI and DI are updated accordingly. CMPS compares two blocks of memory for equality or inequality of the block. It subtracts byte by byte or word by word.

Question No 8

How SCAS instructions can be used to detect Null at the end of a string?

Answer:-

SCAS instruction can be used to search for a 0 in a null terminated string to calculate the length of the string. In this form REPNE will be used to repeat while the null is not there.

Question No 9

How 16-bit MUL operations is different from 32-bit MUL operation?

Answer:-

Multiplication is four bit so the answer is stored in an 8bit register. If the operands were 8bit the answer would be 16bit and if the operands were 16bit the answer would be 32bit. Since eight bits can fit in a byte we have used 4bit multiplication

Question No 10

Write all steps of algorithm for printing number 352.

Answer:-

The steps of our algorithm are outlined below.

- Divide the number by base (10 in case of decimal)
- The remainder is its right most digit
- Convert the digit to its ASCII representation (Add 0x30 to the remainder in case of decimal)
- Save this digit on stack
- If the quotient is non-zero repeat the whole process to get the next digit, otherwise stop
- Pop digits one by one and print on screen left to right

Question No 11

Describe MOVS and CMPS instructions

Answer:-

MOVS

MOVS transfers a byte or word from the source location DS:SI to the destination ES:DI and updates SI and DI to point to the next locations. MOVS is used to move a block of memory. The DF is important in the case of overlapping blocks.

CMPS

CMPS subtracts the source location DS:SI from the destination location ES:DI. Source and Destination are unaffected. SI and DI are updated accordingly. CMPS compares two blocks of memory for equality or inequality of the block. It subtracts byte by byte or word by word. If

used with a REPE or a REPNE prefix is repeats as long as the blocks are same or as long as they are different.

Question No 12

Explain MUL instruction in both cases (i) if the source operand is byte (ii) if the source operand is a word?

Answer:-

MUL (multiply) performs an unsigned multiplication of the source operand and the accumulator. If the source operand is a byte, then it is multiplied by register AL and the double-length result is returned in AH and AL. If the source operand is a word, then it is multiplied by register AX, and the double-length result is returned in registers DX and AX.

Question No 13

IF DF=0 what its represent and IF DF=1 what its represent?

Answer:-

The direction of movement is controlled with the Direction Flag (DF) in the flags register. If this flag is cleared DF=0, the direction is from lower addresses towards higher addresses and if this flag is set DF=1, the direction is from higher addresses to lower addresses. If DF is cleared, DF = 0 this is called the auto increment mode of string instruction, and if DF is set, DF=1, this is called the auto decrement mode. There are two instructions to set and clear the direction flag.

Question No 14

Explain the process of ADC?

Answer:-

The ADC instruction is specifically placed for extending the capability of ADD. Numbers of any size can be added using a proper combination of ADD and ADC. All basic building blocks are provided for the assembly language programmer, and the programmer can extend its capabilities as much as needed by using these fine instructions in appropriate combinations. Further clarifying the operation of ADC, consider an instruction "ADC AX, BX." Normal addition would have just added BX to AX, however ADC first adds the carry flag to AX and then adds BX to AX. Therefore the last carry is also included in the result.

Question No 15

Write a subroutine to clear screen.

Answer:- (Page 93)

; clear screen using string instructions

[org 0x0100]

Jump start

; subroutine to clear the screen

Clrscr: push es

Push ax

Push cx

Push di

Mov ax, 0xb800

Moves, ax ; point es to video base

Xor di, di ; point di to top left column

Mov ax, 0x0720 ; space char in normal attribute

Mov cx, 2000 ; number of screen locations

Cld ; auto increment mode

Rep stosw ; clear the whole screen

Pop di

Pop cx

Pop ax

Pop es

Ret

Start: call clrscr ; call clrscr subroutine

Mov ax, 0x4c00 ; terminate program

Int 0x21

Question No 16

Explain all characteristics of SCAS instruction.

Answer:-

SCAS compares a source byte or word in register AL or AX with the destination string element addressed by ES:DI and updates the flags. DI is updated to point to the next location. SCAS is often used to locate equality or in-equality in a string through the use of an appropriate prefix.

Question No 17

Explain LES and LDS

Answer

The string instructions need source and destination in the form of a segment offset pair. LES and LDS load a segment register and a general purpose register from two consecutive memory locations. LES loads ES while LDS loads DS. Both instructions have two parameters, one is the general purpose register to be loaded and the other is the memory location from which to load these registers. The major application of these instructions is when a subroutine receives a segment offset pair as an argument and the pair is to be loaded in a segment and an offset register.

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